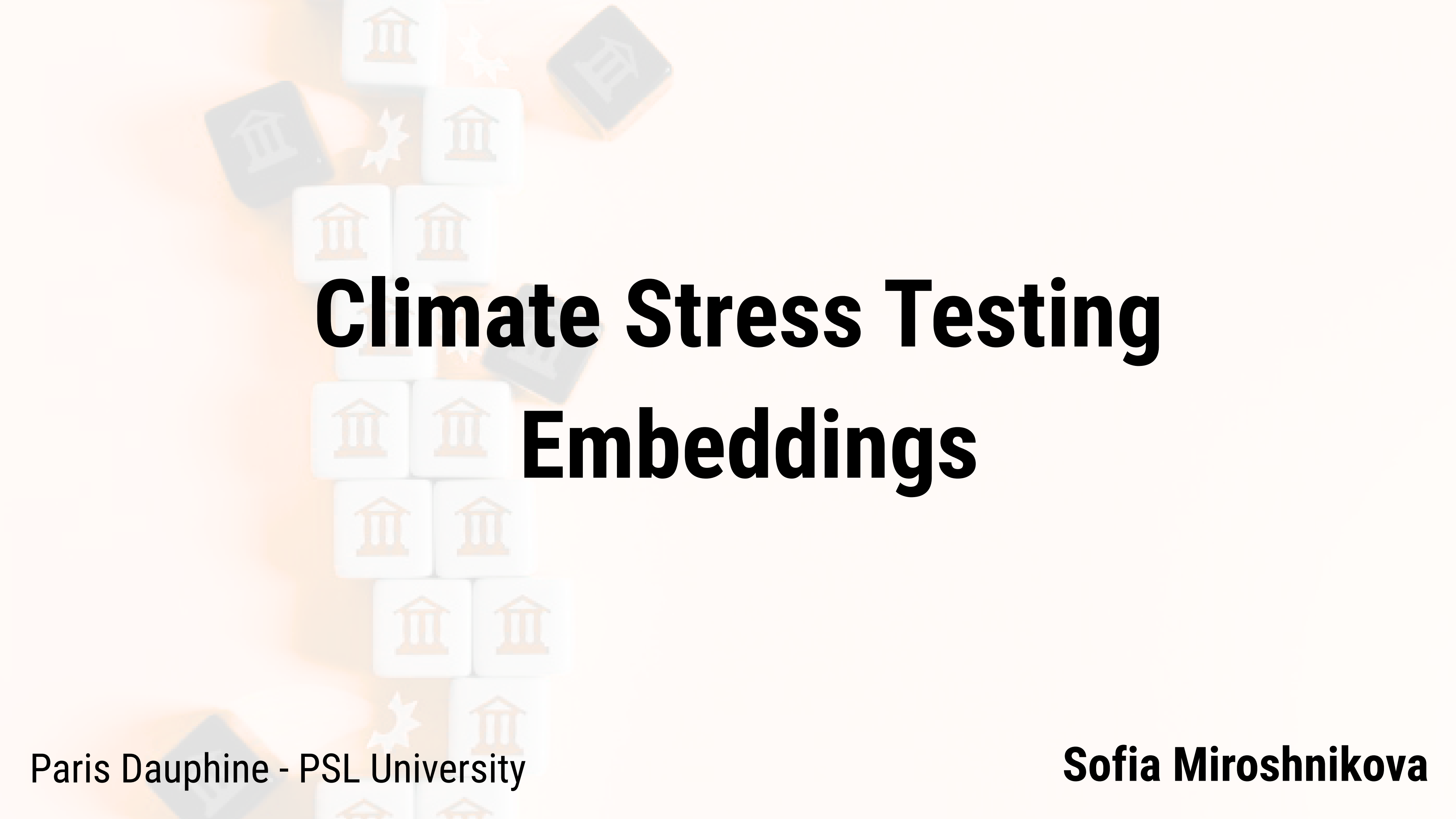


The background features a faint, stylized illustration of a hand holding several blocks. Some blocks are white with an orange university building icon, while others are grey with a white star icon. The overall color palette is warm, with shades of orange and yellow.

Climate Stress Testing Embeddings

A decorative background featuring a hand holding several white and grey blocks. The white blocks are arranged in a grid-like pattern, while the grey blocks are scattered around them. Each block has a stylized icon of a classical building with columns. The hand is positioned on the left side of the image, with fingers visible holding the blocks. The overall color scheme is light and airy, with a soft orange glow emanating from the hand.

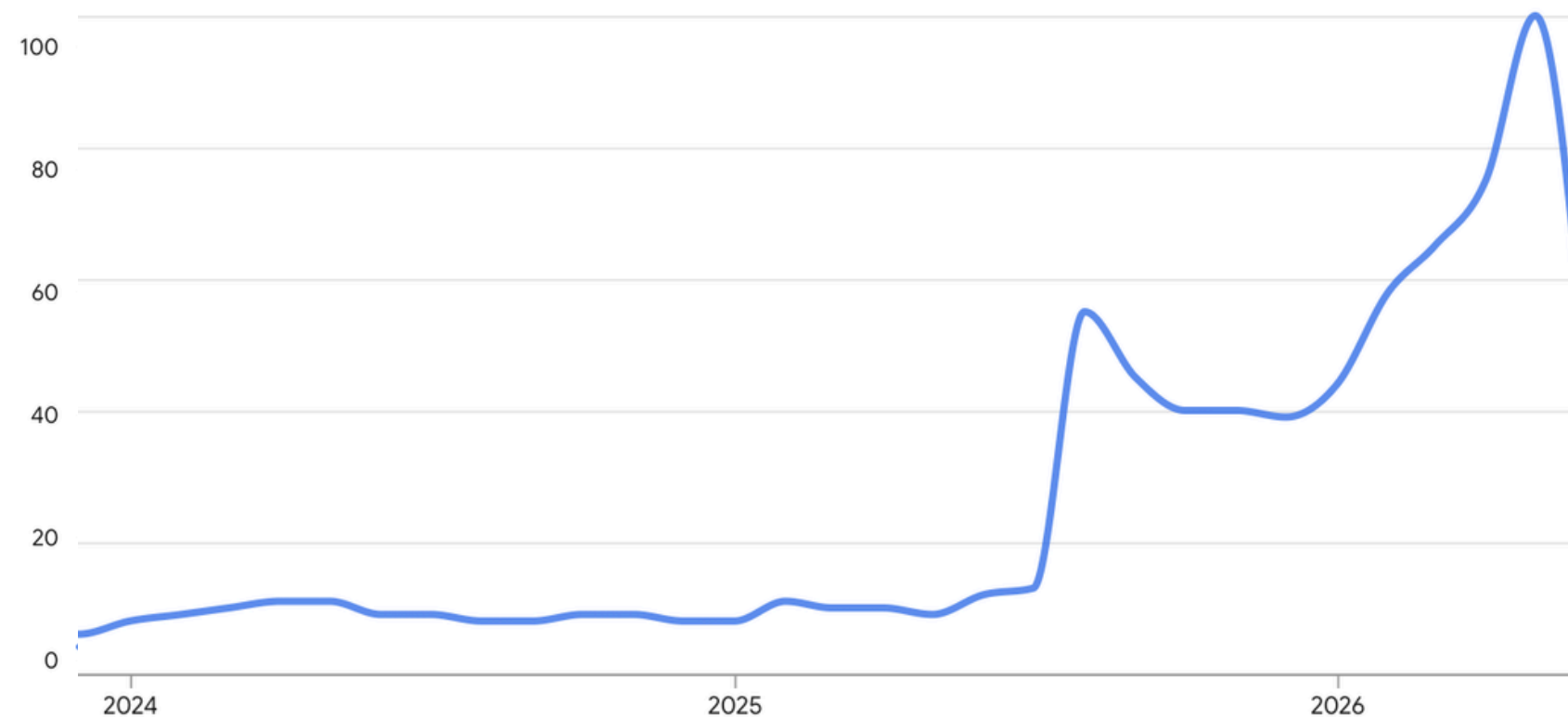
Introduction

The first climate stress tests for European banks have arrived

Interest over time ⓘ

Worldwide · Past 5 years

● embeddings
Search term



APPLE INC, -1.1762833112004325, 9.138016807998637, 4.190403861253175, -1.4575684323794016

Figure 1: Search history of the word “Embeddings” Google Trends

Asset Embeddings*

Xavier Gabaix[†] Ralph S.J. Koijen[‡] Robert J. Richmond[§] Motohiro Yogo[¶]

August 1, 2025

We show theoretically that portfolio holdings contain all relevant information about firms, which can be recovered under empirically realistic conditions. We show empirically that asset embeddings explain these complex economic outcomes much better than firm characteristics.

most similar to United and Zoom. Second, financial institutions, regulators, and central banks could use asset and investor embeddings for risk management and stress testing. They could generate counterfactual changes in asset and investor embeddings, which imply counterfactual changes in asset prices in an asset demand system (Koijen and Yogo, 2019).

The background features a soft-focus image of a hand holding several small, light-colored blocks. Some blocks are white with orange icons of classical buildings, while others are grey with white icons of the same buildings. The text 'Research Question' is overlaid on the right side of the image.

Research Question

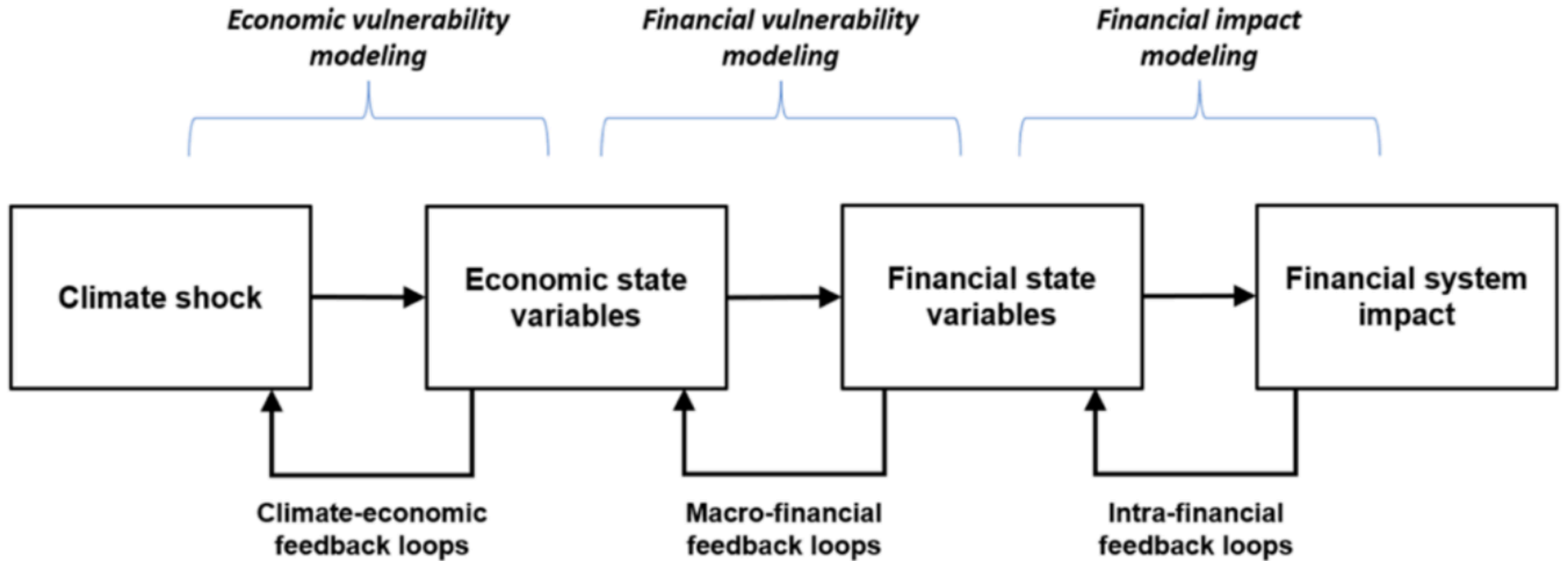
Do embeddings explain some economic outcomes better than firm characteristics and how embeddings can be used in climate stress testing?

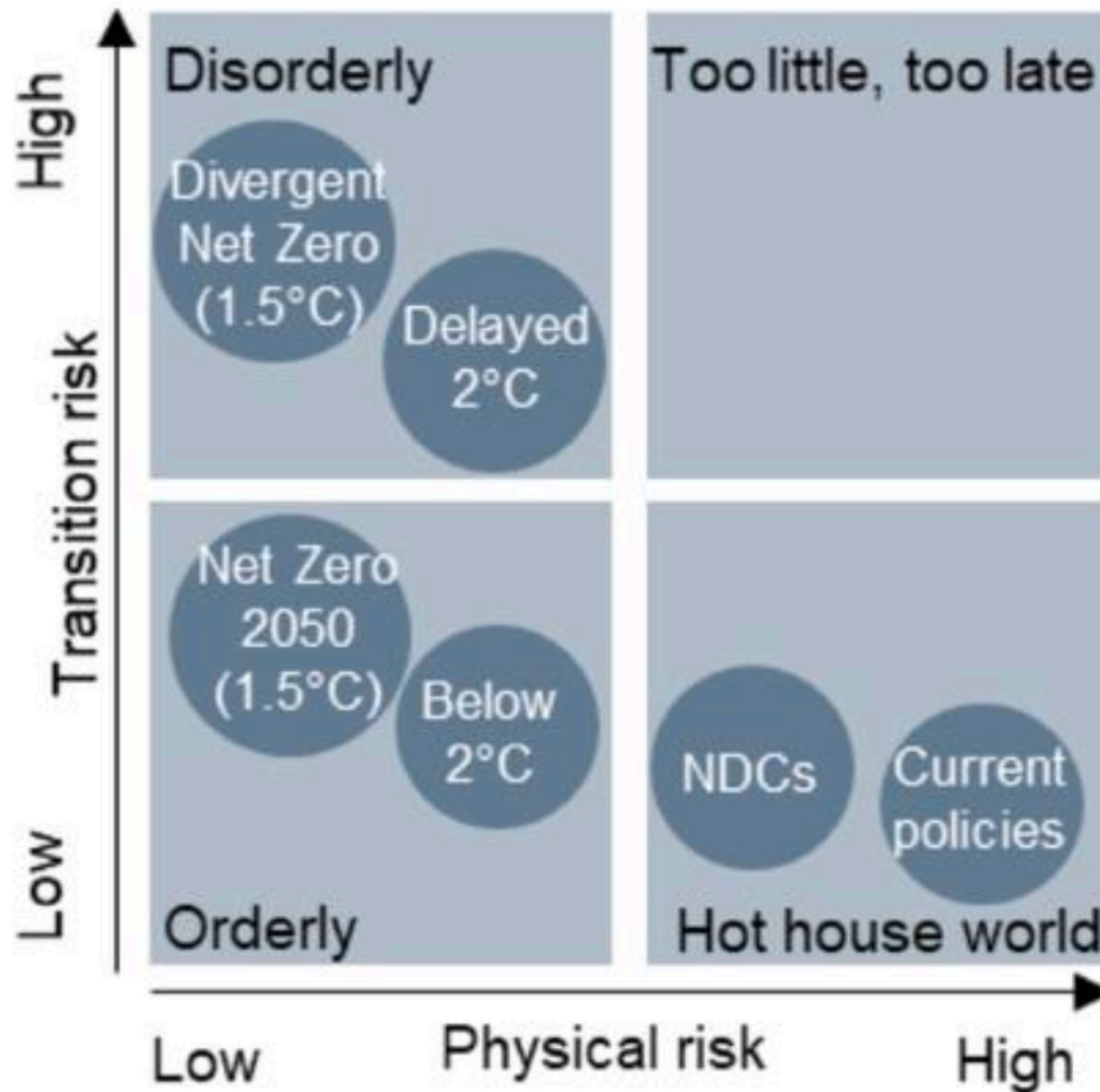
Outcome of the internship:

- Review of climate stress-testing
- Proven that embeddings explain some economic outcomes better than firm characteristics

A background graphic on the left side of the slide shows a hand holding several white and grey blocks. Each block features a stylized icon of a library building with columns. The blocks are arranged in a way that suggests they are being held together, with some blocks slightly offset from others. The overall color scheme is light and airy, with a soft orange glow emanating from the hand area.

Litterature review



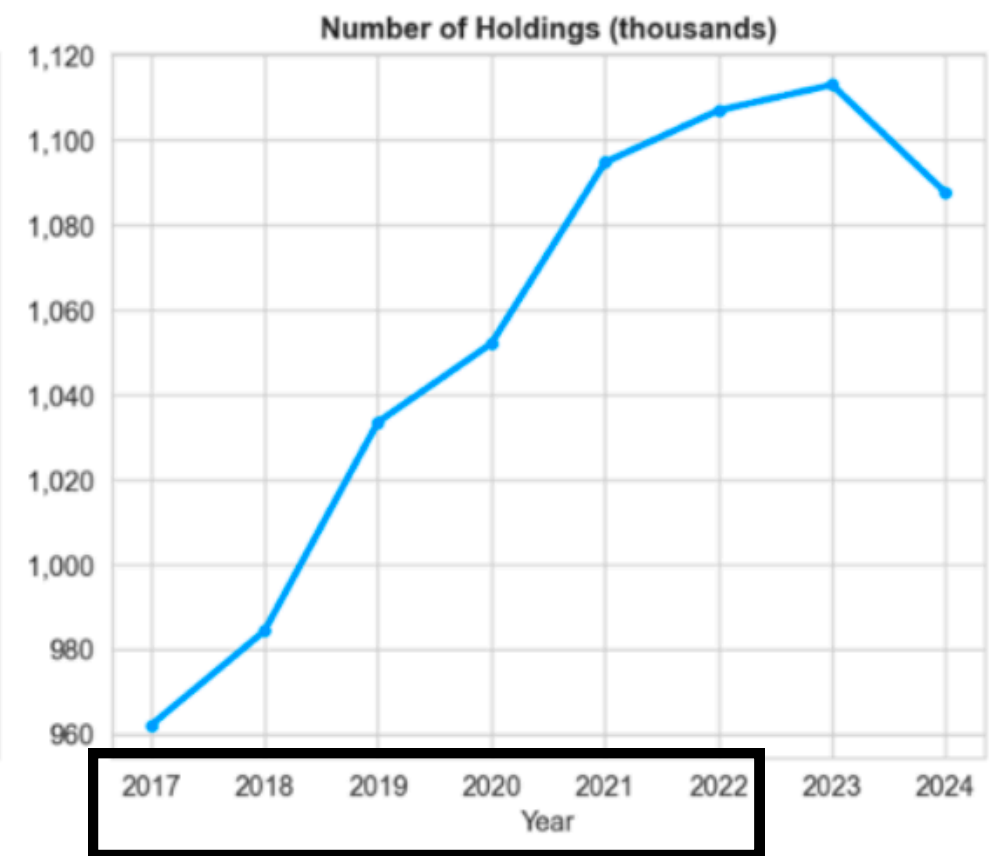
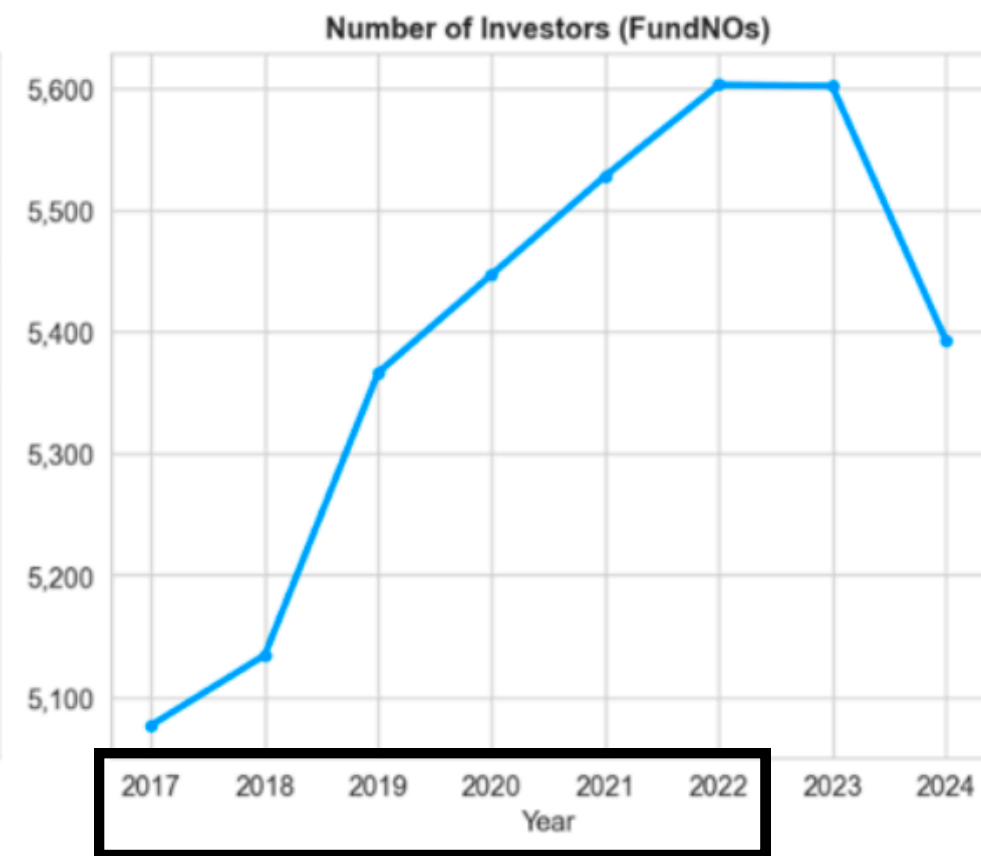
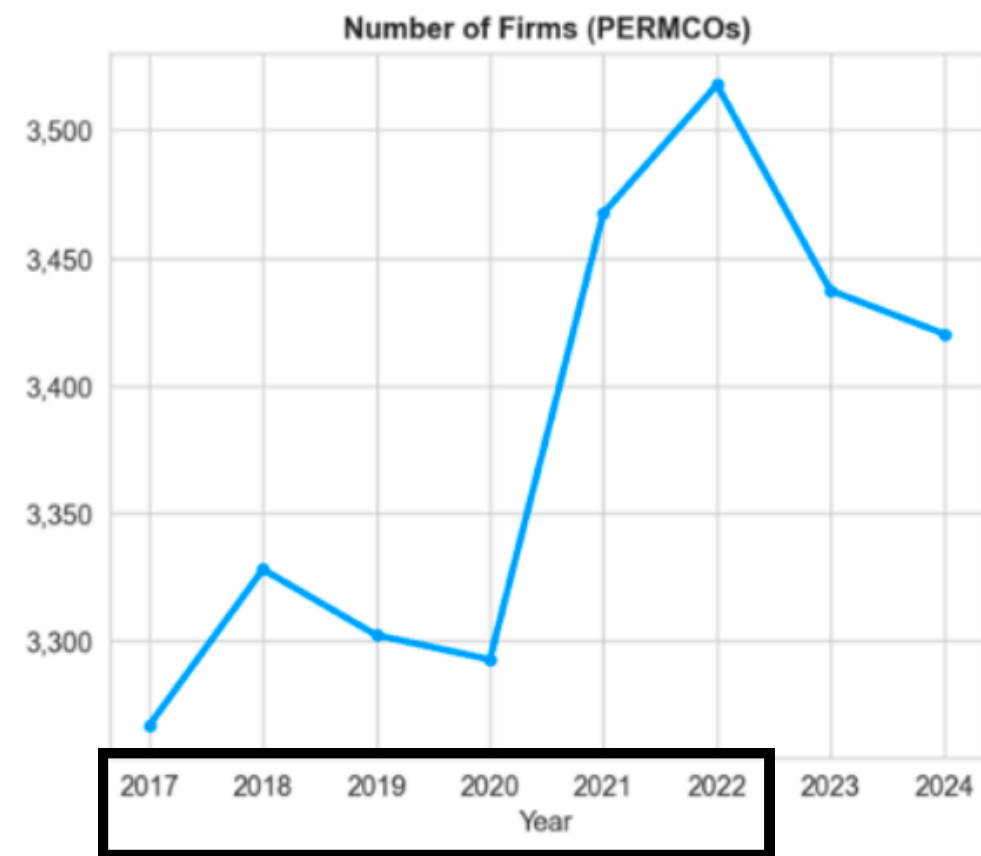
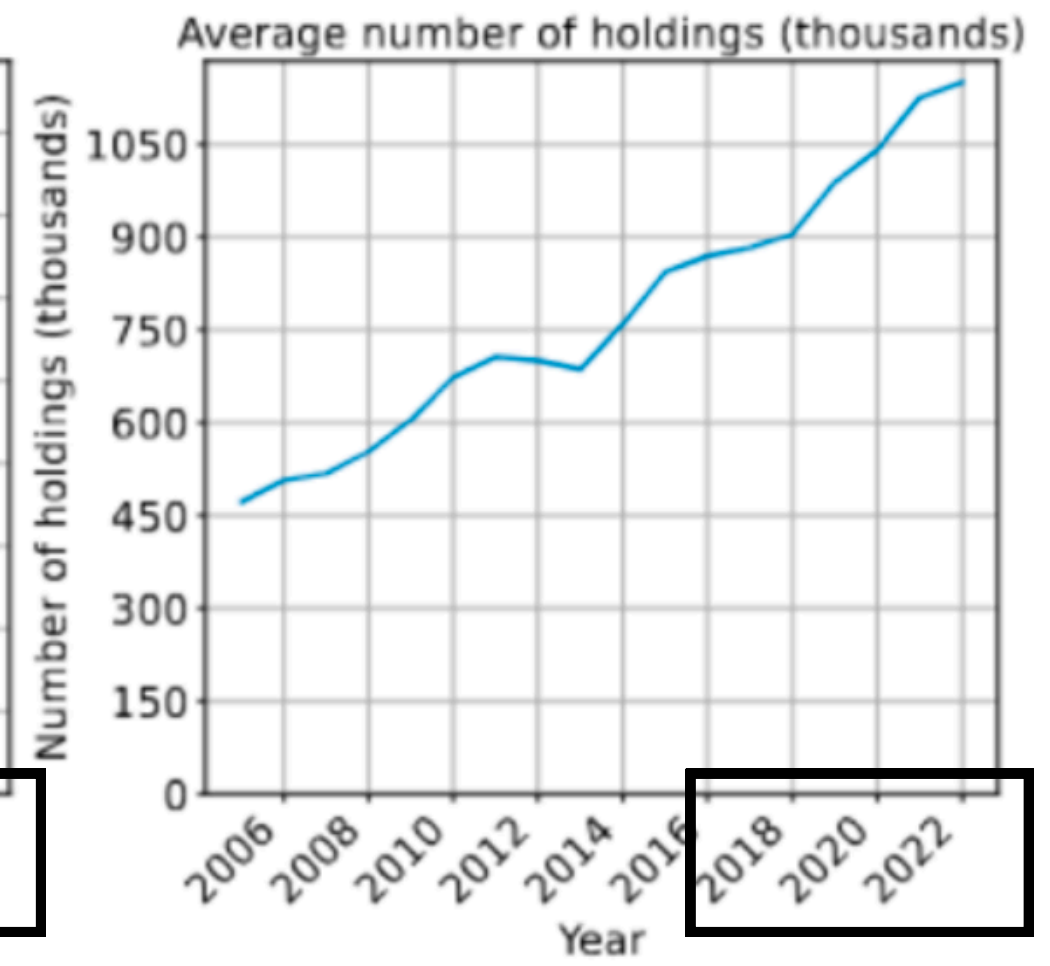
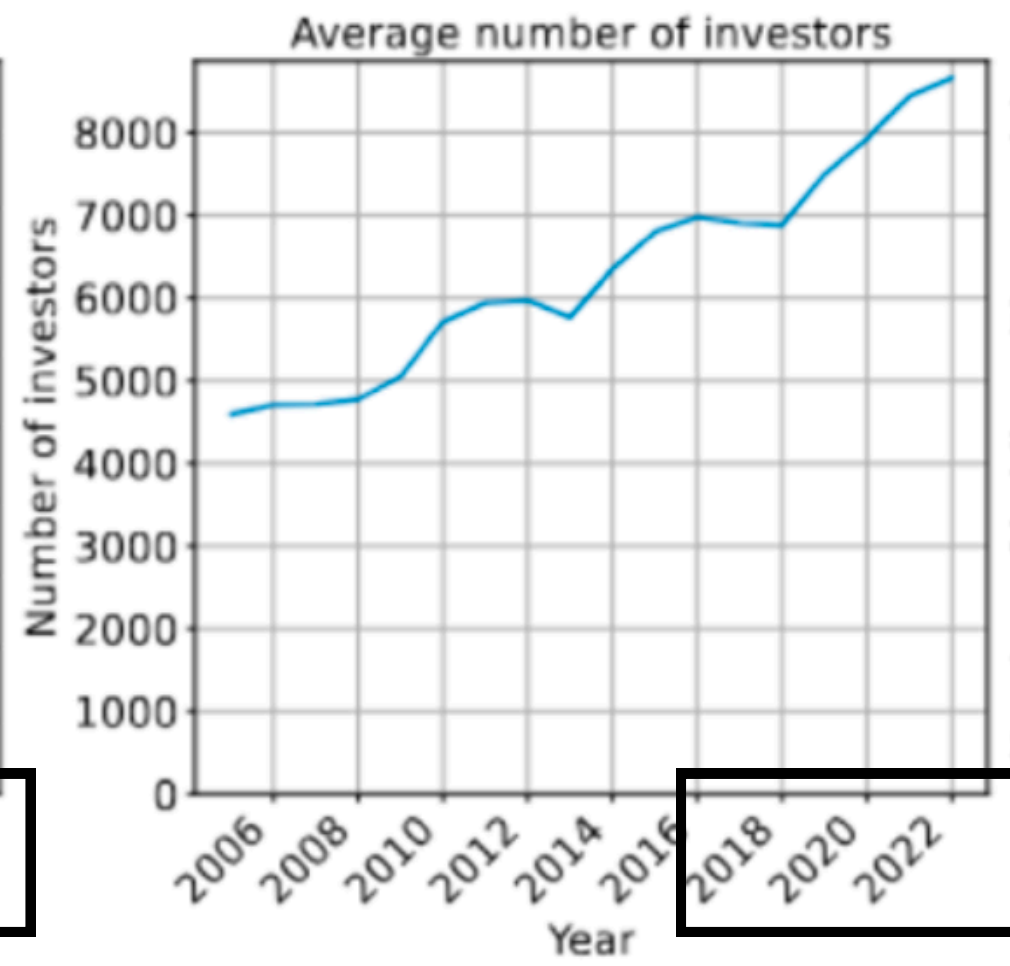
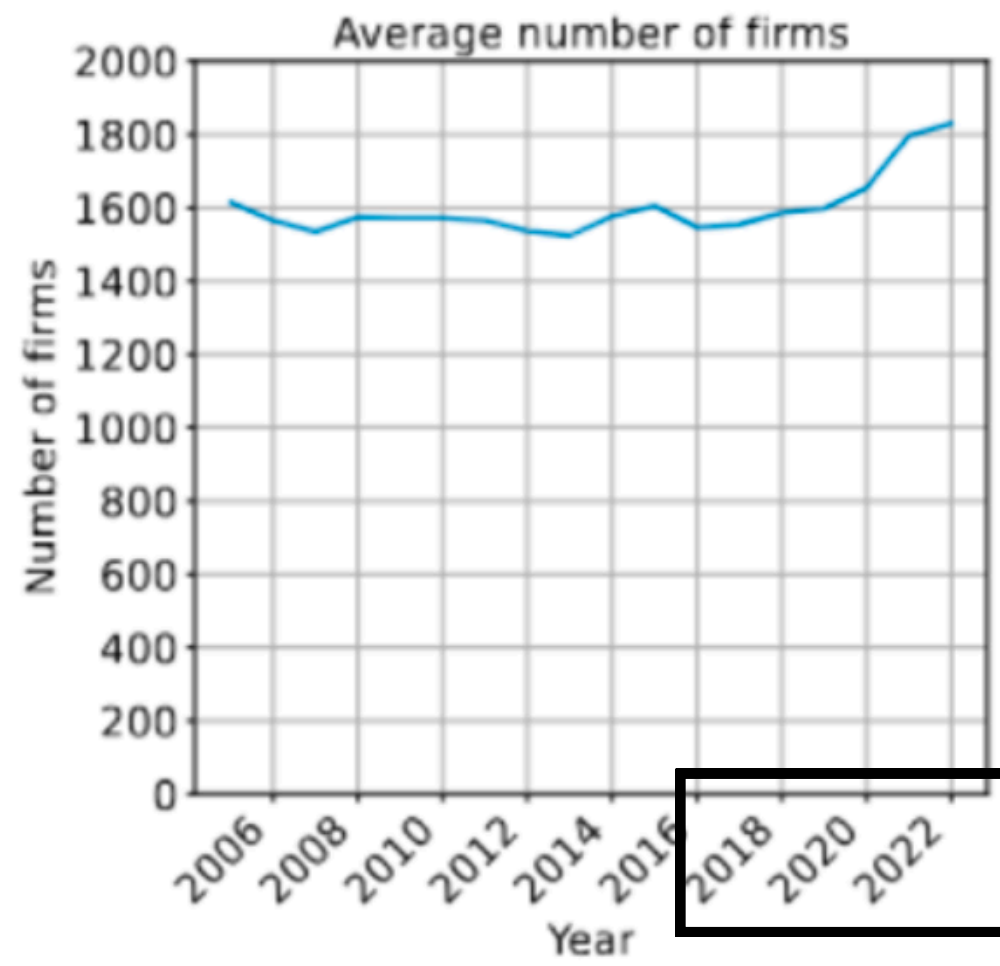


Non-structural approach



A hand is holding a stack of white and grey blocks, each featuring a building icon. The blocks are arranged in a vertical column, with some blocks slightly offset to the left or right. The background is a soft, out-of-focus orange and white gradient.

Data



A decorative graphic on the left side of the slide shows a hand holding several blocks. The blocks are arranged in a vertical column, with some blocks featuring a classical building icon and others featuring a gear icon. The background is a soft, warm gradient.

Methods

Embedding Methods

1. Use of an equation derived by Gabaix et al. (2025)

$$h_{ia} = \lambda_i' x_a + \delta_i + \delta_a + \epsilon_{ia}$$

a) Recommender systems (RS-L-Min)

2. Reliance on the assumption that similar weights are assigned to assets with similar embeddings

b) Shallow Network Models (W2V)

c) Transformer Models (OS-BERT, PS-BERT)

APPLE INC, -1.1762833112004325, 9.138016807998637, 4.190403861253175, -1.4575684323794016

MICROSOFT CORP, -1.1918977509812159, 9.036186482979813, 3.93433060494212, -1.3343132947145113

RIO TINTO PLC, -0.032671000969674784, 0.010955795985696381, -0.06759021035644308, -0.05505026752575304

SHELL PLC, -0.027995742843840862, -0.13957186867351123, -0.05915182649638095, -0.05960355555876016

Benchmark Methods

Relative Valuation Benchmark

Input: Embeddings and data downloaded from WRDS

Step 1 Perform a regression, where residual is the measure we are looking for

Step 2 Divide data 20/80. On 80% ridge regression:

$$p_{at}^{\perp} = \beta'_t x_{at} + \delta_t + \epsilon_{at}$$

Step 3 Compute R-squared

Output: value of R-squared for different embeddings + firm characteristics

Stock Comovement Benchmark

Input: Embeddings and data downloaded from WRDS

Step 1 Divide data 20/80. On 80% ridge regression:

$$r_{am} = \beta_m' x_{a,t-1} + \delta_m + \epsilon_{am}$$

Step 2 Compute R-squared

Output: value of R-squared for different embeddings + firm characteristics

Managed Portfolio Benchmark

Input: Embeddings and data downloaded form WRDS

2	2017Q4,137,AMERICAN INTERNATIONAL GROUP INC,-2.249275413204398,2.077505105774344,0.18237129636311933,0.6980429502309841
3	2017Q4,1543,WEATHERFORD INTERNATIONAL PLC,-0.9677735029575076,-0.6568973651126451,0.9941632457749573,-1.2886508583450327
4	2017Q4,3151,BANK OF AMERICA CORP,-1.7420350944499972,6.530417065008137,2.1371285932442308,0.26535381162143956
5	2017Q4,8045,ORACLE CORP,-1.8251123791605752,5.353262230706208,1.5506217174333774,0.2818509204106605
6	2017Q4,9764,M G M RESORTS INTERNATIONAL,-2.183550514296611,-0.7276533330240016,-0.5370985900310566,0.6627486470688699

Step 1 Calculate prediction probabilities that the masked asset is a certain asset

Step 2 Compute R-squared

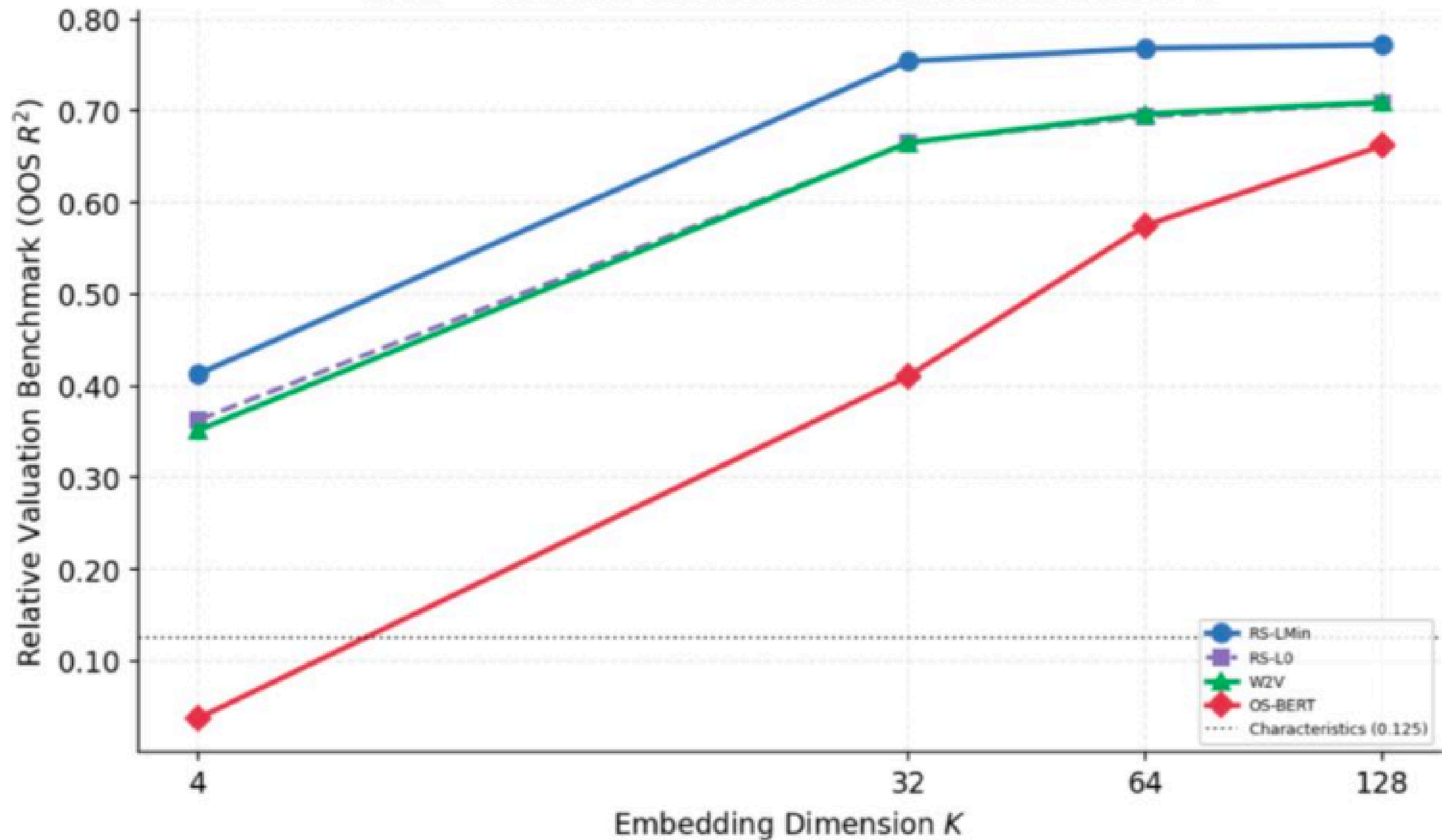
Output: value of R-squared for different embeddings + firm characteristics

A decorative graphic on the left side of the slide shows a hand holding several white and grey blocks. Each block features an icon of a classical building with columns. Some blocks also have a star icon. The blocks are arranged in a way that suggests they are being held together or presented.

Results

	BM1 (RV)	BM2 (RC)	BM3 (MP)
Best model (ours)	RS-L-Min K128	RS-L-Min K128	W2V K128
Best value (ours)	0.772	0.0668	0.398
Characteristics	0.125	0.0625	—
Paper ranking	RS-L-Min > OS-BERT	RS-L-Min > OS-BERT	PS-BERT > W2V
Our ranking	same	same	W2V > PS-BERT

BM1 — Relative Valuation: Performance across K



A hand is shown holding a vertical stack of white, cube-shaped blocks. Each block features a brown icon of a classical building with three columns. Interspersed among these blocks are several white, four-pointed star icons. Surrounding the hand and the stack are several grey, diamond-shaped blocks, each also featuring the same brown building icon. The background is a soft, out-of-focus light orange or peach color.

Discussion

	BM1 (RV)	BM2 (RC)	BM3 (MP)
Best model (ours)	RS-L-Min K128	RS-L-Min K128	W2V K128
Best value (ours)	0.772	0.0668	0.398
Characteristics	0.125	0.0625	—
Paper ranking	RS-L-Min > OS-BERT	RS-L-Min > OS-BERT	PS-BERT > W2V
Our ranking	same	same	W2V > PS-BERT

Differences in results: Differences in the dataset coverage

Next steps:

1. Vector green-brown investor
2. Stress-testing with the framework by (Koijen and Yogo, 2019)

ESG:

Score difference

Limits that could be encountered further on:

Economic interpretability of embeddings

What benefits?

1. Work with API and major databases
2. Literature review
3. Attendance of the conferences